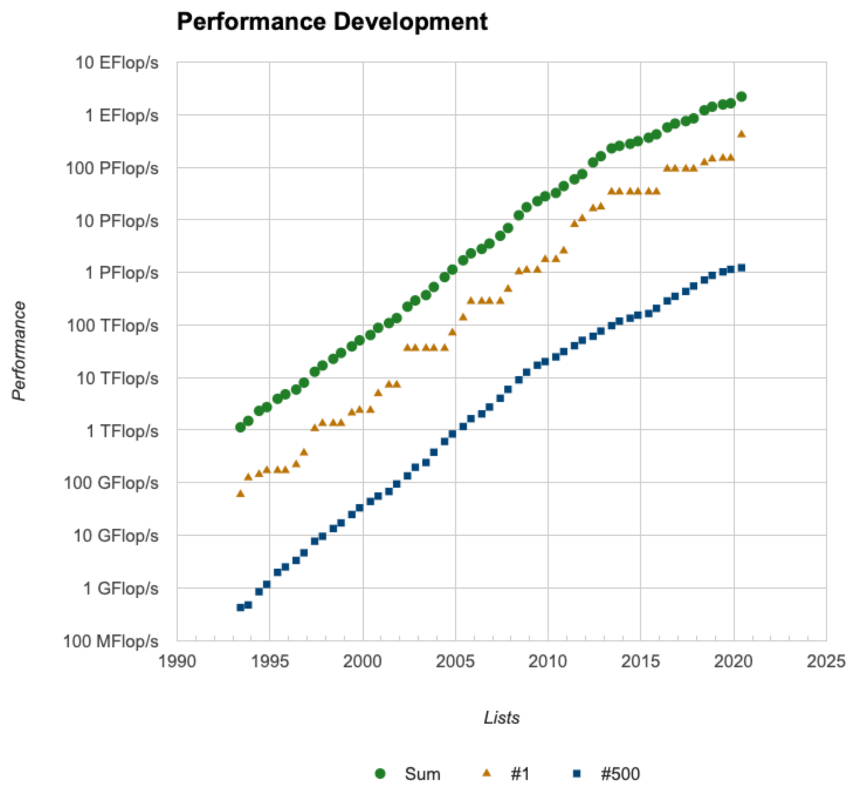
Lecture Unit 9.4

Linear algebra benchmarks

Benchmarks

- Linear algebra workloads are often used for benchmarking
- The "Top 500" list ranks HPC systems according to measured performance on the Linpack benchmark



Linpack

- Linpack measures **Fl**oating point **op**erations per **s**econd (Flops/s) with a dense linear algebra workload:

  - Solves **Ax**=**b** (where **b** and **x** are vectors and **A** is a dense NxN matrix) using lower-upper (LU) factorisation

  - This is a dense matrix operation with algorithmic intensity ~O(N) and is compute-bound

High Performance Linpack (HPL)

- High Performance Linpack (HPL) is used in Top500 list
- It is permitted to change the size of the matrices so that floating point performance is close to the peak value
- Measure actual HPL performance (R_{max}) usually a significant fraction of the theoretical maximum performance (R_{peak})

November 2024 Top 4:

Best measured HPL performance

Theoretical maximum performance

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	**El Capitan** - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,039,616	1,742.00	2,746.38	29,581
2	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	**Aurora** - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	**Eagle** - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	

63% peak

66% peak

High Performance Conjugate Gradients (HPCG)

- HPL is very floating point intensive but we are also interested in other types of applications
- High Performance Conjugate Gradients (HPCG) is representative of sparse linear algebra workloads
- Compliments HPL as it is measuring something different
- Solves 3D Poisson's equation using a 27-point stencil
- Finds an iterative solution to **Ax**=**b** where **A** is a sparse matrix

High Performance Conjugate Gradients (HPCG)

- Performs O(N) floating point operations
- Requires O(N) transfers from memory
- Lower algorithmic intensity than HPL
- HPCG performance (flop/s) much lower than HPL → small fraction of peak performance

HPCG November 2024 Top 4:

Best measured HPL performance

Rank	TOP500 Rank	System	Cores	Rmax (PFlop/s)	HPCG (TFlop/s)
1	6	**Supercomputer Fugaku** - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	16004.50
2	2	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	14054.00
3	3	**Aurora** - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	5612.60
4	8	**LUMI** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	4586.95

2.97% peak

0.68% peak

Unit summary

- High Performance Linpack (HPL) and High Performance Conjugate Gradients (HPCG) are two complimentary HPC benchmarks
- HPL measures performance on a dense linear algebra workload and achieves a significant fraction of peak performance
- HPCG measures performance on a sparse linear algebra workload and the performance achieve is much less than the peak value
- For more information see SAB 4.4 (HPL) and SAB 4.6 (HPCG)